

Modular Boundary Responses for Unknot Recognition

Certified Euler obstructions, singular-safe compression,
and deterministic threshold refinement

ProveIt research continuation
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8 October 2026

Abstract

We continue the implementation program in *ProveIt/Topology/UnknotRecognition*. The maintained marked-continuation observer reconstructs closed diagrams and evaluates integer signed-Laplacian determinants. A prior research prototype reuses a rational boundary response. We give and implement a finite-field continuation of that prototype, retaining singular interior directions and reusing the actual suffix geometry. Its auxiliary primes reduce integer Euler characteristics; the underlying Khovanov complex remains over $\mathbb{F}_2$.

Three quantitative results organize the improvement. First, a checkerboard suffix with $b > 0$ frontier darts has exactly $b/2$ black open face fragments, so every potentially nonzero residual determinant has dimension at most $b - 2$. Second, modular residues of whole completed components give sound, monotone lower bounds on reduced Khovanov rank. Retaining the exact Euler sum yields a closed formula for the strongest ℓ^1 lower bound with those data. Third, if each component coordinate is bounded by C_a , accumulated modulus $M > \max_a C_a$ already makes the threshold-one test exact; unique coefficient reconstruction is unnecessary. We prove explicit bounds on C_a and identify the distinction between this completion theorem and the finite prime palette implemented by the prototype.

Fresh-setup paired measurements show a 26.67-fold improvement over the maintained direct evaluator and a 3.97-fold improvement over the earlier rational prototype on a 200-query workload with 47 genuine interior vertices. The small-interior cases and all seven raw recognition examples show regressions. The new backend therefore remains optional. No general quasi-polynomial running-time bound for the maintained recognizer is proved. The package includes proofs, source changes, independent arithmetic and geometric audits, replay checks, full timing samples, and a targeted research agenda.

Research status. This is an AI-assisted working paper, cross-checked with separate arithmetic routines and exact small-instance computations, not a refereed publication. Established results, repository-derived constructions, proved refinements, experiments, and open obligations are distinguished throughout. The source baseline is recorded exactly in the accompanying provenance file.

Contents

1	Scope, source audit, and principal conclusions	3
1.1	The question addressed	3
1.2	What this continuation adds	3
1.3	Current literature and the complexity target	4
1.4	Reading the empirical conclusion	4
2	Marked completion and the integer Euler obstruction	5
2.1	Relative complexes and a persistent marked point	5
2.2	Four integer coordinates	6
2.3	Recovering the vector from two scalar evaluations	6
2.4	The exact evaluation at one	7
2.5	The signed determinant and its phase	7
2.6	Relative shifts and multiplicity saturation	7
3	The exact boundary size and the modular response kernel	8
3.1	Cut-face fragments	8
3.2	A precise count instead of a loose linear estimate	8
3.3	Boundary-only geometric acceptance	8
3.4	The signed Laplacian quotient	9
3.5	Elimination over each prime separately	9
3.6	What is retained and what is rebuilt	10
4	Modular lower bounds and deterministic threshold completion	11
4.1	A residue is information about an integer	11
4.2	An explicit solution to the integer norm problem	11
4.3	Threshold recovery requires less than coefficient recovery	12
4.4	A useful limitation of the exact-sum enhancement	13
4.5	How many individual primes can miss an existing obstruction?	13
5	Coefficient bounds and the remaining complexity barrier	14
5.1	An explicit bound for an actual completed suffix	14
5.2	How many bits of modulus suffice?	15
5.3	A cost account for the implemented observer	15
5.4	What a width bound would and would not buy	16
6	Implementation, replay, and correctness evidence	17
6.1	An optional backend with a small integration surface	17
6.2	Saturation and transactional resource behavior	18
6.3	What the replay verifier proves	18
6.4	Independent arithmetic and actual-diagram audits	18
6.5	Full-suite status and an inherited limitation	19
7	Experiments: arithmetic gains and recognition regressions	20
7.1	Protocol and comparison targets	20
7.2	A controlled family of actual query streams	20
7.3	The repeated-query improvement	20
7.4	The full scans are slower on this corpus	21
7.5	The supported performance conclusion	23
8	Further research questions and proposed priorities	23
8.1	Near-term implementation and experimental questions	23
8.2	Questions that address the quasi-polynomial barrier	24
8.3	External algorithm audits and integration milestones	25

1 Scope, source audit, and principal conclusions

1.1 The question addressed

The practical question is whether repeated observations of a partially computed knot complex can become sufficiently cheap to justify earlier recognition. The asymptotic question is whether such observations can help the program avoid the exponentially large intermediate representations that obstruct a general quasi-polynomial bound. These questions require different evidence. A fast determinant subroutine answers the first only on workloads where it is used often enough. It answers the second only when accompanied by control of the number and size of the complexes that must be constructed before it succeeds.

The source examined here is the public ProveIt revision

8a95834940cf77cdab1b39571ffc102ca8b6bede.

All references to the “maintained” implementation mean that pinned revision, before this package’s additions. The repository already contains 28 numbered reports, numerous integrated improvements, and a Python implementation far beyond the earliest scanning baseline. Its relevant files are `fast/fastunknot/shadow_scan.py`, `fast/fastunknot/integer_determinant.py`, `fast/fastunknot/tait_blocks.py`, and the research prototype `fast/determinant_research/boundary_tait.py`. The maintained explanations are `synthesis/determinant_continuations.tex`, `synthesis/boundary_reuse.tex`, `synthesis/tait_block_s.tex`, and `synthesis/shadow_fallback.tex` [11].

Report 26 already supplies the marked four-residue obstruction and a singular-safe rational terminal kernel. Later maintained work supplies cut-face fragments, a boundary-only geometric acceptance check, articulation factorization of direct cofactors, and a budgeted Euler fallback. We use those results openly. Reintroducing a Schur complement or an Euler norm without this attribution would misstate the continuation’s contribution.

1.2 What this continuation adds

The deliverable has four connected parts.

1. An exact terminal count, $t = b/2$, and the resulting $b - 2$ residual dimension bound for a nonempty frontier. This sharpens the earlier loose linear bound for the actual cut-face construction.
2. A modular observer with a fully proved integer interpretation. It combines congruences by the Chinese remainder theorem (CRT), retains the exact coordinate sum, and solves a four-dimensional integer norm minimization explicitly.
3. A deterministic threshold completion theorem. The modulus required to decide whether the existing bound exceeds one is smaller than the modulus required to reconstruct all its integer coordinates.
4. An optional implementation in the maintained package, with independent replay of its positive observations and paired comparisons against both relevant predecessors.

The basic modular-determinant idea is established numerical algebra [5]. Boundary response matrices also have a substantial literature [4]. Our claims concern the explicit proofs and implementation given here; a limited literature search is not a certification of historical priority for every elementary lemma.

Theorem 1.1 (Combined observer guarantee). *Suppose a validated classical knot diagram has a proper scan prefix whose relative complex has been obtained by the algebraically valid operations implemented by the maintained scanner. Let C_a be its whole differential components, with positive multiplicities m_a , and let $x_a \in \mathbb{Z}^4$ be the marked completed Euler vectors. For a product M of distinct validated odd primes, the algorithm below computes a number L_M satisfying*

$$0 \leq L_M \leq B_4 := \sum_a m_a \|x_a\|_1 \leq \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

*It may therefore return **KNOTTED** whenever $L_M > 1$. If $|x_{a,r}| \leq C_a$ and $M > \max_a C_a$, then*

$$L_M > 1 \iff B_4 > 1.$$

For a suffix with $b > 0$ frontier darts, each modular determinant query after preparation uses a matrix of dimension at most $b - 2$, or returns the proved zero residue without forming that matrix. These statements remain valid at primes where the interior matrix becomes more singular.

The proof is distributed through sections 2 to 4. The hypothesis about the relative complex is essential: a randomly chosen matrix satisfying $d^2 = 0$ is not enough. Likewise, $L_M \leq 1$ is not an unknot certificate, even when the threshold comparison with B_4 is complete.

1.3 Current literature and the complexity target

Lackenby’s July 2026 preprint develops an explicit hierarchy framework and efficiently encoded certificates [7]. Proposition 9.1 bounds the number of iterations by

$$L(g + 1)^L,$$

where L is a hierarchy-length bound and g a pattern-complexity bound. The refined algorithm controls hierarchy length using handle complexity; Theorem 14.1 establishes, under its manifold and handle-structure hypotheses, the existence of an essential hierarchy with a polynomial-size encoding and polynomial-time verification. The paper leaves the proposed speed-up to further work. These are substantial algorithmic foundations, but polynomial certificate verification does not supply a polynomial or quasi-polynomial certificate-construction time.

The exact exponent also deserves care. The March 2021 Oxford technical handout states $2^{O((\log N)^3)} = N^{O((\log N)^2)}$ [8]. The Oxford seminar abstract instead states $N^{O(\log N)}$ [9]. Both are quasi-polynomial, but the latter is stronger. We keep the repository’s stronger target visible while using $2^{(\log N)^{O(1)}}$ when discussing the broad class. No result here converts one announced exponent into another.

A separate September 2026 preprint by Musick explicitly claims polynomial unknot recognition [10]. This changes what a current literature note must acknowledge. The present work does not validate or integrate that claimed algorithm. A limited review identifies the grammar-to-geometric-local-minimum step and total compressed-description accounting as obligations for a separate audit; it supplies no counterexample or disproof. Our definite complexity claim is about the maintained ProveIt implementation: it still has no established general quasi-polynomial guarantee.

The representation obstacle is concrete. In their January 2026 preprint, Kelomäki and Schütz prove exponential behavior for basis-oriented scanning and divide-and-conquer on particular positive or alternating three-braids, while also obtaining polynomial-time structural formulas for three-braid Khovanov homology [6]. Thus a small frontier and a large explicit homology basis can coexist. A succinct answer may nevertheless be tractable. This continuation compresses an observation of the complex; it does not prove that the complex itself remains succinct.

1.4 Reading the empirical conclusion

On the favorable repeated-query workload, 47 interior vertices are eliminated once and 200 actual boundary matchings are observed. The modular path is 26.67 times faster than the maintained direct path and 3.97

times faster than the rational prototype. At 127 interior vertices and 50 queries, the corresponding ratios are 10.74 and 10.22.

Conversely, the all-terminal workload slows down, the smallest nontrivial interior favors the rational prototype, and the raw recognizer examples all favor the existing direct observer. These measurements justify an optional backend and further scheduling research. They do not justify changing the default recognizer or claiming a general recognition speed-up.

Claim category	Status in this package
Modular arithmetic soundness	Proved, implemented, independently audited.
Exact terminal count	Proved for the checked cut-face representation.
Deterministic threshold completion	Proved; implementation reports whether its finite palette suffices.
Repeated-query acceleration	Measured with setup included and both predecessors compared.
Whole-recognizer acceleration	Not demonstrated on the tested corpus.
General quasi-polynomial recognition	Not established by this work.

2 Marked completion and the integer Euler obstruction

2.1 Relative complexes and a persistent marked point

Fix a classical one-component planar diagram D with N crossings and a validated permutation of its crossings. A proper prefix leaves a suffix with $n \geq 1$ crossings. The scanning construction represents the prefix by a complex in the dotted cobordism category. Each object is a boundary matching with its homological and quantum shifts; closed circles have been removed through the standard delooping operations. Differentials, cancellations, and their homotopies are the actual maps supplied by this construction [1, 2].

Choose a point on an edge incident to the final unprocessed crossing. It remains in the suffix at every proper-prefix observation. On completion, the marked circle carries the action of $A = \mathbb{F}_2[x]/(x^2)$. The marked quotient is applied after the actual suffix has been attached.

Lemma 2.1 (Why completion and reduction preserve the contractions). *Suppose the prefix complex is decomposed and simplified by chain maps and chain homotopies that commute with the marked action after completion. Applying the marked quotient preserves every recorded chain-homotopy equivalence.*

Proof. If maps f, g and a homotopy h satisfy $fg - 1 = dh + hd$, all maps are A -linear. Their induced maps on the quotient satisfy the same equality, because passage to the quotient respects composition and addition. The other homotopy identity is treated identically. This argument does not require the quotient functor to be flat or exact on arbitrary short exact sequences. $\square$

Let $\mathcal{C}_a$ be a whole connected component of the differential graph after such simplification. The absence of differentials between these groups gives a direct sum of complexes. Completing with the actual suffix and applying the marked quotient gives

$$\widetilde{\text{Kh}}(K; \mathbb{F}_2) \cong \bigoplus_a H\left(\widetilde{\text{cl}}(\mathcal{C}_a)\right)^{\oplus m_a}, \quad (1)$$

up to recorded shifts, whose effect on the norms below is harmless. Shared copies can have different global shifts; each has the same norm.

2.2 Four integer coordinates

For a finite bigraded complex X , define its residue-four Euler vector by

$$\sigma_r(X) = \sum_{h, j \equiv r \pmod{4}} (-1)^h \dim_{\mathbb{F}_2} X^{h,j}, \quad 0 \leq r < 4. \quad (2)$$

Differentials preserve the quantum grading, so taking Euler characteristics after homology leaves this vector unchanged. A homological shift negates all four coordinates when the shift is odd. A quantum shift cyclically permutes them. Both preserve the ℓ^1 norm.

Proposition 2.2 (Existing marked-component bound). *With the provenance of lemma 2.1, put $x_a = \sigma(\widetilde{\text{cl}}(\mathcal{C}_a))$. Then*

$$B_4 = \sum_a m_a \|x_a\|_1 \leq \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

In particular, $B_4 > 1$ certifies that K is nontrivial.

Proof. For each residue, let e_r, o_r be the total dimensions of the even and odd homology groups. Then $|e_r - o_r| \leq e_r + o_r$. Summing over residues and using equation (1) proves the inequality. Reduced Khovanov rank one detects the unknot; the maintained $\mathbb{F}_2$ criterion follows from the standard unknot-detection theorem and change-of-coefficients implications [3]. The rank-one implication is the existing recognition premise, not a new assertion about the auxiliary prime fields used below. $\square$

The order of operations is part of the theorem. For each whole component, sum all signed, shifted object vectors first. Then take a norm, and finally sum with positive multiplicities. A pair of object vectors u and $-u$ may belong to one contractible complex. Their true Euler contribution is zero, while $\|u\|_1 + \|-u\|_1$ is positive. Neither modular arithmetic nor early stopping permits taking a norm midway through this sum. Stopping after a *completed* component is safe because all remaining component norms are nonnegative.

2.3 Recovering the vector from two scalar evaluations

We use the repository's raw reduced polynomial convention

$$J_D(q) = \sum_{s \in \{0,1\}^n} (-q)^{|s|} (q + q^{-1})^{c(s)-1}. \quad (3)$$

Here $c(s)$ counts the circles of the smoothing. The completion contains no explicit crossing-free circles outside this diagram; previously delooped circles are represented by the prefix's shifted objects and multiplicities.

Changing a smoothing changes the circle count by one in parity. Hence all quantum degrees in equation (3) have parity $\epsilon = (c(0) - 1) \bmod 2$. If

$$a = J_D(1), \quad d = i^{-\epsilon} J_D(i) \in \mathbb{Z},$$

the only potentially nonzero coordinates are

$$\sigma_\epsilon = \frac{a + d}{2}, \quad \sigma_{\epsilon+2} = \frac{a - d}{2}. \quad (4)$$

Indeed, evaluation at one adds the two residue sums, while evaluation at i subtracts them after removing the common factor i^ϵ . In particular,

$$\|\sigma(D)\|_1 = \max(|a|, |d|). \quad (5)$$

This identity will give small and explicit coefficient bounds.

2.4 The exact evaluation at one

Let c be the number of link components and $c_0 = c(0)$ the all-zero circle count. In the raw convention,

$$a = (-1)^{n+c+c_0} 2^{c-1}. \quad (6)$$

One way to see the sign is to choose an orientation and write h for the number of one-smoothings in its Seifert state and s for its Seifert circles. Circle parity gives $s \equiv c_0 + h \pmod{2}$. The orientable Seifert surface has $s - n \equiv c \pmod{2}$. Thus $h \equiv n + c + c_0 \pmod{2}$. The normalized reduced evaluation at one is 2^{c-1} , and the raw shift supplies the sign above.

The existing implementation computes the same integer by independently orienting the completed link. A reusable boundary connectivity summary can instead determine c and c_0 without revisiting the suffix. Both versions remain available as cross-checks. The integer 2^{c-1} has $O(n)$ bits; a boundary-sized combinatorial traversal is not automatically a boundary-only bit-cost bound for constructing this integer.

2.5 The signed determinant and its phase

For a connected completed projection, checkerboard color its regions. At a crossing e , let b_e be the smoothing bit that does not join its black sectors, and put $w_e = (-1)^{b_e}$. Set $B = \sum_e b_e$. If G is the signed black Tait graph with v vertices, then

$$J_D(i) = (-i)^{B+v-1} \tau(G), \quad \tau(G) = \det L_G^{(0)}, \quad (7)$$

where $L_G^{(0)}$ is a grounded Laplacian cofactor.

At $q = i$, all states except one-circle states vanish. Those states correspond to spanning trees of the Tait graph. Their phases give equation (7); the matrix-tree expansion is polynomial in the edge weights, so the signs do not require a positive Laplacian. Loops contribute to B , although their incidence vectors vanish in the Laplacian. Removing them from the phase would invalidate the observation. If the projection is disconnected, every smoothing has more than one circle and $J_D(i) = 0$.

For connected accepted geometry we also have

$$B + v - 1 \equiv \epsilon \pmod{2}.$$

Choose any unsigned spanning tree. Its one-circle state has smoothing-weight parity $B + v - 1$, while circle parity gives $|s| \equiv c_0 - 1 \pmod{2}$. This proves the identity even when the signed tree sum cancels. It is a combinatorial parity condition, not merely a check that the computed Gaussian integer lies on the correct axis. The implementation checks it even when a modular determinant vanishes, so a zero residue cannot conceal a phase inconsistency.

2.6 Relative shifts and multiplicity saturation

The maintained grading recovery assigns a relative quantum shift t_o to every object in a component, checking every nonzero differential and cycle. For a map of dot degree d between matchings with m arcs and c glued circles, its equation is $t_{\text{target}} - t_{\text{source}} = m - c + 2d$. Object o contributes $(-1)^{h_o} \text{rot}_{t_o} \sigma(D_o)$. Only the shift modulo four is needed after consistency has been verified.

For threshold T , replacing a positive multiplicity m by $\min(m, T + 1)$ preserves whether $m \|x\|_1 > T$, because the norm is an integer. The same statement extends to sums of nonnegative terms. The maintained scanner uses cap three; this remains sound for our threshold one. It does not permit saturating signed Euler coefficients before aggregation.

Finally, a proper-prefix restriction remains mandatory. At the closed terminal stage, separate unmarked scalar generators are not automatically free marked A -summands. The modular observer refuses that stage. An UNKNOT verdict comes only from the completed rank-two unreduced calculation.

3 The exact boundary size and the modular response kernel

3.1 Cut-face fragments

At a suffix stage, retain its $4n$ crossing darts. Let ρ rotate each crossing counterclockwise and let α pair darts joined by retained internal edges. Boundary darts are the b darts where this partial involution is undefined. The partial face map is

$$f(d) = \rho(\alpha(d)).$$

It is an injective partial map. Its components are directed cycles and directed paths, including paths consisting of one dart. They are the closed face fragments and open face fragments of the suffix.

A checkerboard coloring of the original diagram assigns a color to each dart corner. Internal face-map steps preserve color. Rotating to the next corner changes color. Therefore each fragment has one well-defined inherited color. The common Tait graph has black fragments as vertices and one signed edge for each retained crossing. Open black fragments are its terminals; closed black fragments are its interior vertices. Keeping fragments is necessary: removing the prefix may split an original region, so merely remembering its original region identifier would not encode every completion correctly.

If boundary labels a, b are paired, with darts d_a, d_b , the two newly inserted face connections are

$$d_a \longrightarrow \rho(d_b), \quad d_b \longrightarrow \rho(d_a). \quad (8)$$

Thus completion only identifies open fragments. Closed fragments remain distinct. These are exactly terminal identifications of the common graph.

3.2 A precise count instead of a loose linear estimate

Theorem 3.1 (Half the frontier is the terminal set). *For $b > 0$, a checkerboard-colored suffix has exactly b open face fragments, of which exactly $b/2$ are black and $b/2$ are white. In particular, the common Tait graph has exactly*

$$t = b/2$$

terminals, whichever shading is chosen.

Proof. There are b missing outgoing ends of f , precisely the boundary darts d . There are b missing incoming ends, precisely the darts $\rho(d)$ for those same boundary darts. Every noncycle component of an injective partial map on a finite set is a directed path with one missing incoming end and one missing outgoing end. Hence the number of paths is exactly b .

Let O_B, I_B denote the black missing-outgoing and missing-incoming counts, and define O_W, I_W similarly. Each path is monochromatic, giving $O_B = I_B$ and $O_W = I_W$. Rotation changes color, so the correspondence $d \mapsto \rho(d)$ gives $I_B = O_W$ and $I_W = O_B$. All four quantities are therefore $b/2$. The black paths are exactly the black open fragments. An isolated dart is a length-zero path and has the same one-incoming, one-outgoing deficiency; it causes no exception. $\square$

The earlier boundary-reuse account only required a safe $O(b)$ terminal bound. The exact equality gives a sharper storage contract and, after elimination, a useful explicit matrix-size bound. It does not require the entire frontier to lie on one disk boundary, and it makes no Catalan-state assumption.

When $b = 0$, there are no open fragments. The code chooses one black vertex as an artificial grounding terminal. This case has $t = 1$ and must not be substituted into a formula involving $b - 2$.

3.3 Boundary-only geometric acceptance

The common graph is valid for a query only after three checks. First, the matching must cover every boundary label exactly once. Second, every connection in equation (8) must preserve the inherited color. Third, the completed rotation system must be spherical.

Let F be the number of completed faces and k the number of completed projection components. Store the numbers of closed fragments and closed projection components during preparation. The query uses disjoint-set unions only among open fragments and among projection components touching the boundary. It then checks

$$F = n + 2k. \quad (9)$$

The completed ribbon graph has $V = n$, $E = 2n$. Its total Euler characteristic is $F - n = 2k - 2 \sum_i g_i$, where $g_i \geq 0$ are its orientable genera. Thus equation (9) is equivalent to every component having genus zero. There is no need to reconstruct all suffix faces for each query.

Both local pairings used by the Euler observer, namely opposite crossing darts for link strands and adjacent darts for the zero smoothing, also admit path summaries. Gluing either summary to the query matching computes c and c_0 in boundary-sized work. The acceptance check and these two counts together give the exact phase and the exact value at one.

An inherited-color incompatibility declines this representation; it does not prove knottedness. A malformed or nonspherical query is rejected as invalid input to this oracle. In the public recognition pipeline all queried matchings come from a validated classical scan, and a declined optional representation causes fallback.

3.4 The signed Laplacian quotient

For an edge $e = uv$ with integer weight w_e , add $w_e(e_u - e_v)(e_u - e_v)^\top$ to the Laplacian. This definition includes parallel edges and gives zero for a loop. It yields an integer symmetric matrix with zero row sums, without a positivity assumption.

Order the interior and terminal vertices to write

$$L = \begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}.$$

For a terminal partition with $q + 1$ blocks, ground its first block. Let Q be the $t \times q$ matrix indicating the other blocks. Grounding the quotient graph gives the matrix

$$L_Q = \begin{pmatrix} A & BQ \\ Q^\top B^\top & Q^\top D Q \end{pmatrix}. \quad (10)$$

Every interior vertex remains separate. All terminal identifications are encoded by the columns of Q . This identity follows by applying the identification map to each edge incidence vector, so it remains valid if some edges become loops or cancel by opposite weights.

3.5 Elimination over each prime separately

Fix an odd prime p . We reduce the original integer L modulo p and eliminate only interior coordinates by symmetric pivots. If the remaining interior block has a nonzero diagonal entry, use a one-coordinate pivot. If all its diagonals vanish but an off-diagonal entry u is nonzero, use

$$P = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad \det P = -u^2 \neq 0.$$

If neither exists, the remaining interior block is zero. Symmetry ensures that these cases exhaust the possibilities.

For an invertible pivot block P , replace the remaining block by its Schur complement and multiply an accumulated factor δ by $\det P$. Apply the same operations to interior-terminal couplings and terminal-terminal entries. Simultaneous row and column permutations contribute the square of a sign, so they do not change the determinant. No rational pivot profile is reused at another prime.

Theorem 3.2 (Singular-safe modular response). *Let $r = \text{nullity}_{\mathbb{F}_p} A$. The preparation produces $\delta \in \mathbb{F}_p^\times$, $R \in \mathbb{F}_p^{r \times t}$, and a symmetric $S \in \mathbb{F}_p^{t \times t}$ such that for every terminal partition,*

$$\det L_Q = \delta \det \begin{pmatrix} 0_r & RQ \\ Q^\top R^\top & Q^\top S Q \end{pmatrix} \quad \text{in } \mathbb{F}_p. \quad (11)$$

If $r > q$, the answer is zero. If $r \leq q$, the residual dimension is at most $2(t-1)$. In particular, for $b > 0$ it is at most $b-2$.

Proof. At each pivot the block determinant identity proves the equality between the old determinant and the pivot determinant times the new Schur-complement determinant. Because only interior coordinates are eliminated, these identities commute with multiplication of the coupling and terminal blocks by Q and $Q^\top$. Induction over all pivots yields equation (11). The remaining interior block is zero and has dimension $\text{nullity } A$ because every extracted pivot is nonsingular.

If $r > q$, the linear map RQ has a nonzero left null vector. Extending that vector by zeros on the terminal coordinates gives a null vector of the residual symmetric matrix. Otherwise its dimension is $r + q \leq 2q \leq 2(t-1)$. Apply theorem 3.1 for the last assertion. $\square$

When $r = q$, a useful independent identity is

$$\det L_Q = (-1)^r \delta \det(RQ)^2. \quad (12)$$

It follows by swapping the two block column groups and obtaining a block triangular matrix. Its determinant is independent of S . This is a field identity; an ordering or real sign must not be assigned to its modular value.

Example 3.3 (A singular modular interior with a nonzero answer). Take a triangle whose edge weights are 1, 2, 1, and designate two vertices as terminals. With the remaining vertex first,

$$L = \begin{pmatrix} 3 & -1 & -2 \\ -1 & 2 & -1 \\ -2 & -1 & 3 \end{pmatrix}, \quad \tau(G) = 5.$$

Over $\mathbb{Q}$, the interior block [3] is invertible. Over $\mathbb{F}_3$ it is zero, so a rational Schur complement cannot simply be reduced using its denominator. The modular response retains one radical direction; for the unmerged terminal partition its residual matrix has off-diagonal entry 1 and determinant $-1 \equiv 2 \pmod{3}$, exactly $5 \pmod{3}$. At $p = 5$, the final determinant is zero modulo five although the integer determinant is nonzero. Neither phenomenon is an error or a reason to infer an integer zero.

3.6 What is retained and what is rebuilt

The implementation keeps one active suffix geometry. For each requested prime it builds one response, then caches cofactors by canonical terminal partition. Two different boundary matchings with the same black partition share a determinant. They still undergo their own geometry checks and may have different link counts; these data are not inferred from the partition alone.

If $r > t-1$, every query residue is zero and the kernel can discard its unused large residual arrays. Otherwise it retains at most an $r \times t$ coupling and a $t \times t$ response, with $r \leq t-1$. Changing the suffix discards these active matrices. The separately cached completed values remain exact for their full keys, which include the stage, matching, and prime.

Preparation is dense and still costs $O(v^3)$ field operations for a common graph with v vertices. The gain is that this cost is paid once per prime and suffix, while subsequent determinants are boundary-sized. The theorem does not guarantee that a scan asks enough queries to repay this setup.

4 Modular lower bounds and deterministic threshold completion

4.1 A residue is information about an integer

The complex remains over $\mathbb{F}_2$. Its graded dimensions and Euler sums are integers. We use an auxiliary prime p only to compute those integers modulo p , evaluating equation (4) with $2^{-1} \in \mathbb{F}_p$. The Gaussian phase is represented by one of $(1, 0), (0, -1), (-1, 0), (0, 1)$; there is no need to adjoin a square root of minus one to $\mathbb{F}_p$.

For a whole component, add the shifted object residues with their homological signs modulo p . Applying CRT to this final four-vector reconstructs its congruence class modulo the product of the processed primes. This ordering requires only four accumulated integers per component. Reconstructing every object's full integer determinant first is unnecessary.

Definition 4.1 (Balanced and sum-constrained norms). For odd $M \geq 3$, let $\text{bal}_M(z)$ be the unique representative of $z \bmod M$ in $[-(M-1)/2, (M-1)/2]$. For a residue vector $z \in (\mathbb{Z}/M\mathbb{Z})^d$, set

$$D_M(z) = \sum_j |\text{bal}_M(z_j)|.$$

If an exact integer sum a is known and $a \equiv \sum_j z_j \pmod{M}$, set

$$F_M(z, a) = \min \left\{ \|y\|_1 : y \in \mathbb{Z}^d, y \equiv z \pmod{M}, \sum_j y_j = a \right\}.$$

For our application, $d = 4$ and $a_a = \sum_r x_{a,r}$ is the exact completed component Euler value, obtained from the already available evaluations at one. The two modular bounds are

$$B_M = \sum_a m_a D_M(x_a), \quad L_M = \sum_a m_a F_M(x_a, a_a). \quad (13)$$

Proposition 4.2 (Soundness and monotone refinement). *For every odd M ,*

$$B_M \leq L_M \leq B_4.$$

If $M \mid M'$, then $B_M \leq B_{M'}$ and $L_M \leq L_{M'}$. Also $L_M \geq \sum_a m_a |a_a|$.

Proof. Balanced representatives minimize each coordinate's absolute value among all its integer lifts. Requiring the exact sum can only raise that minimum. The true integer vector is feasible for the sum-constrained problem, proving the upper bound. A congruence class modulo M' is a subset of its congruence class modulo M , so both minima are monotone. Finally the triangle inequality gives $\|y\|_1 \geq |\sum_j y_j| = |a_a|$ for every feasible vector. $\square$

Each rejection is therefore deterministic and sound, even when the first prime is chosen heuristically. Our implementation uses a fixed default prime, not probabilistic stabilization of reconstructed integers.

4.2 An explicit solution to the integer norm problem

Theorem 4.3 (Minimum norm with a known sum). *Put $b_j = \text{bal}_M(z_j)$ and $h = (a - \sum_j b_j)/M$. If $h = 0$, then $F_M(z, a) = \sum_j |b_j|$. Otherwise let $u_1 \geq \dots \geq u_s > 0$ be the absolute values of those b_j whose sign is opposite to h . Then*

$$F_M(z, a) = \sum_j |b_j| + M|h| - 2 \sum_{j=1}^{\min(|h|, s)} u_j. \quad (14)$$

A minimizing lift can be constructed using $O(d \log d)$ comparisons and integer arithmetic, independent of the magnitude of h except for bit cost.

Proof. Write a lift as $y = b + Mw$, with $w \in \mathbb{Z}^d$ and $\sum_j w_j = h$. For one positive step in coordinate j , the change in absolute value is M unless $b_j < 0$, in which case the first step costs $M - 2|b_j|$. Every further positive step costs M . Negative steps have the analogous first-step discount when $b_j > 0$. Since M is odd and $|b_j| \leq (M - 1)/2$, all first-step costs are strictly positive.

If some $w_i > 0$ and some $w_j < 0$, moving both one step toward zero leaves $\sum w$ unchanged and strictly decreases the objective. Thus a minimizer has all nonzero w_j of the sign of h . Allocate the $|h|$ required steps to the largest available first-step discounts. Once each selected coordinate has received one step, every remaining marginal cost is M . This proves equation (14). Sorting the at most d discounts and putting any remaining steps into one already correctly signed coordinate constructs an attaining lift. No loop of length $|h|$ is necessary. $\square$

For example, modulo three the residues $(1, 1)$ have balanced norm two. With exact sum five, a minimizing lift is $(4, 1)$ and its norm is five. Feasibility alone would not prove a lower bound: an arbitrary lift may have norm larger than the true integer vector. The optimization theorem is what makes the returned minimum safe.

4.3 Threshold recovery requires less than coefficient recovery

Let $T \geq 0$ be an integer threshold and suppose certified coordinate bounds $|x_{a,r}| \leq C_a$ are available. They need not be equal across components.

Theorem 4.4 (Balanced threshold completion). *If*

$$M > \max_a (C_a + \lfloor T/m_a \rfloor),$$

then $B_M > T$ if and only if $B_4 > T$.

Proof. One implication follows from soundness. If $B_M \leq T$, every balanced coordinate in component a has absolute value at most $\lfloor T/m_a \rfloor$. Any nonzero difference between the true coordinate and that representative is a multiple of M . It would force the true coordinate to have magnitude at least $M - \lfloor T/m_a \rfloor > C_a$. Thus all true coordinates equal their balanced representatives and $B_4 = B_M \leq T$. $\square$

Theorem 4.5 (Sharper completion with the exact sum). *If*

$$M > \max_a (C_a + \lfloor T/(2m_a) \rfloor), \tag{15}$$

then $L_M > T$ if and only if $B_4 > T$. In particular, at $T = 1$,

$$M > \max_a C_a \tag{16}$$

suffices.

Proof. Again soundness gives one implication. Suppose $L_M \leq T$ and choose a minimizing lift y_a for each component. Then $\|y_a\|_1 \leq \lfloor T/m_a \rfloor$. If $x_a \neq y_a$, the vector $(x_a - y_a)/M$ is a nonzero integer vector with coordinate sum zero. It therefore has a coordinate at least one and another at most minus one. Among the corresponding two coordinates of y_a , at least one has absolute value at most $\lfloor T/(2m_a) \rfloor$. On that coordinate,

$$|x_{a,r}| \geq M - |y_{a,r}| \geq M - \lfloor T/(2m_a) \rfloor > C_a,$$

a contradiction. Thus every $x_a = y_a$, proving $B_4 = L_M \leq T$. $\square$

At threshold one this avoids the usual $M > 2 \max C_a$ requirement for unique centered reconstruction. It decides a Boolean comparison, not every large coordinate. The distinction is useful even when the modulus is not large enough to identify a general integer vector uniquely.

Example 4.6 (Strictness and fixed-palette blindness). For any odd $M \geq 3$, the vector $(M, -M, 0, 0)$ has exact sum zero and is indistinguishable from zero modulo M . With $C = M$, condition equation (16) fails at equality, and the modular norm is zero while the true norm is $2M$. Thus strict inequality is necessary for a theorem based only on coordinate bounds. More generally, if the product of all primes in a fixed palette is P , the vector $(P, -P, 0, 0)$ is invisible to the entire palette. These are algebraic counterexamples to an overstrong inference; no claim is made that a prescribed vector is realized by a knot prefix.

The general bound is likewise sharp. For one component of weight m , let $s = \lfloor T/(2m) \rfloor$, choose odd $M > 2s + 1$, and take $y = (-s, s)$, $x = (M - s, -M + s)$ with $C = M - s$. They have the same exact sum and the same residues. The small lift has weighted norm $2ms \leq T$, whereas the large lift can exceed T . Equality at $M = C + s$ therefore cannot replace the strict comparison.

4.4 A useful limitation of the exact-sum enhancement

Proposition 4.7 (No extra threshold-one rejection after the Euler check). *Suppose $M \geq 3$ is odd and $\sum_a m_a |a_a| \leq 1$. Then*

$$B_M > 1 \iff L_M > 1.$$

Proof. If $B_M \leq 1$, each component's balanced vector has norm at most one and $|a_a| \leq 1$. The difference between a_a and the balanced coordinate sum is divisible by M and has absolute value at most two. It is therefore zero. Every balanced vector already satisfies the exact sum, so $L_M = B_M \leq 1$. The other implication follows from $B_M \leq L_M$. $\square$

This prevents an attractive but incorrect performance claim. In the actual recognizer, the exact Euler obstruction is checked first. The sum-constrained formula does not then discover additional threshold-one rejections beyond balanced residues. Its value here is the stronger completion criterion and an explicit general higher-threshold interface. We implement it for that certified interpretation, not as an unmeasured source of extra knot detections.

4.5 How many individual primes can miss an existing obstruction?

CRT is one route to completion; a bound on individual bad primes is another useful planning tool.

Proposition 4.8 (Prime-exposure bound). *Assume $|x_{a,r}| \leq H$ and $B_4 > 1$. In a pool of distinct odd primes each at least $\ell > 1$, either every prime detects $B_p > 1$, or the number of inconclusive primes is at most (in this second case $H \geq 2$)*

$$\left\lfloor \frac{\log(H(H^2 - 1))}{\log \ell} \right\rfloor < \frac{3 \log(H + 1)}{\log \ell}.$$

Proof. If every coordinate is 0 or ± 1 , every odd prime preserves each absolute value, and every prime detects the obstruction. Otherwise choose one coordinate x with $|x| \geq 2$. An inconclusive prime has $B_p \leq 1$, so this coordinate is congruent to 0 or ± 1 . Hence it divides the nonzero integer $x(x - 1)(x + 1)$. The product of k distinct such primes is at least ℓ^k and at most $H(H^2 - 1)$, proving the bound. $\square$

Uniformly random ordering of a certified prime pool consequently admits an expected exposure estimate, although no randomness is needed for soundness. We do not implement such a scheduler. The result concerns exposing an *already present* component obstruction; it neither guarantees that one exists nor bounds the work required to reach the corresponding prefix.

5 Coefficient bounds and the remaining complexity barrier

5.1 An explicit bound for an actual completed suffix

The completion theorem is useful only with a proved size bound. Let $n \geq 1$ be the number of suffix crossings. A completed suffix is obtained by pairing its frontier labels, with no additional crossing-free circles supplied from outside. Circles already removed by delooping in the prefix are represented by the relative complex and its shifts; they must not be silently added to this closed diagram.

Lemma 5.1 (Crossings bound the number of link components). *For a spherical n -crossing diagram in which every link component meets a crossing, the number c of link components satisfies $c \leq n$.*

Proof. For component i , let s_i be its number of self-crossings and u_i the number of mixed-crossing incidences on it. If $s_i > 0$, then $s_i + u_i/2 \geq 1$. If $s_i = 0$, the component is a simple closed curve on the sphere. Its number of intersections with every other closed curve is even, by mod-two intersection on the sphere. Because it meets a crossing, u_i is positive and at least two. Thus $s_i + u_i/2 \geq 1$ in this case as well. Summing over components counts each self-crossing once and each mixed crossing twice with coefficient one half, so $c \leq \sum_i (s_i + u_i/2) = n$. $\square$

Every link component of our completed suffix meets a retained crossing: each frontier endpoint belongs to a retained crossing, and the matching has no closed component of its own. Hence the lemma applies even if the projection is disconnected.

Proposition 5.2 (Per-object and per-component coordinate bounds). *Let $\sigma \in \mathbb{Z}^4$ be the marked four-residue Euler vector of a completed n -crossing suffix object, and put $a = J(1)$. Then*

$$|a| \leq 2^{n-1}, \quad |d| \leq 2^{n-1}, \quad (17)$$

$$\|\sigma\|_1 \leq 2^{n-1}, \quad |\sigma_r| \leq \left\lfloor \frac{|a| + 2^{n-1}}{2} \right\rfloor. \quad (18)$$

For a whole differential component containing s_a represented objects, with object Euler values a_o , a valid bound is

$$C_a = \sum_{o \in \mathcal{C}_a} \left\lfloor \frac{|a_o| + 2^{n-1}}{2} \right\rfloor \leq s_a 2^{n-1}. \quad (19)$$

Proof. The value-at-one formula gives $|a| = 2^{c-1} \leq 2^{n-1}$. For a connected projection, the signed Tait cofactor is a sum of products of ± 1 over spanning trees. If its Tait graph has v vertices, every tree chooses exactly $v - 1$ of the n crossing edges, so its absolute value is at most $\binom{n}{v-1}$. For $n \geq 1$, each binomial coefficient is at most 2^{n-1} : it is a term in either the even or the odd binomial sum, and both sums equal 2^{n-1} . Loops and parallel edges do not invalidate this argument. For a disconnected projection, the value at i is zero, so the same bound holds. Multiplication by a Gaussian unit does not change $|d|$.

The two nonzero possible coordinates are $(a + d)/2$ and $(a - d)/2$; therefore their total absolute value is $\max(|a|, |d|)$, and each absolute value has the stated integer bound. Homological signs and quantum shifts only negate or permute coordinates. Summing over a whole component and using the triangle inequality gives equation (19). $\square$

This deliberately elementary bound is inexpensive to compute and is valid for signed, nonalternating diagrams. A smaller certified bound can only improve the stopping criterion. The implementation uses the first expression in equation (19), because the exact a_o values have already been computed by the Euler prefilter.

5.2 How many bits of modulus suffice?

Let $s_{\max} = \max_a s_a$. At threshold one, a product with

$$M > s_{\max} 2^{n-1}$$

is sufficient. Thus the sufficient threshold has $n + \log_2 s_{\max} + O(1)$ bits. A small-prime product can overshoot that target by the bit length of its last prime; this does not affect the asymptotic bound for the Boolean comparison. No exponential-size integer *bit length* is hidden in this step, although $s_{\max}$ itself may count exponentially many represented objects.

For a supply of distinct primes each of at least w bits, the number needed is at most

$$O\left(1 + \frac{n + \log(s_{\max} + 1)}{w}\right),$$

provided enough such primes are supplied and validated. This is a statement about the arithmetic interface, not a claim that a fixed machine-word palette works for arbitrarily large inputs. An asymptotic implementation can enlarge the prime range and include the cost of finding and certifying the primes. For completeness, let $L \geq 1$ be the required modulus bit length. Sieve odd primes up to a cutoff X , doubling X until their product exceeds the certified bound. The classical Chebyshev estimate $\vartheta(X) = \sum_{p \leq X} \log p = \Theta(X)$ [12] shows that this stops with $X = O(L)$. Sieving, exact product construction, and primality certification by the sieve therefore take polynomial time and space in L , and each selected prime has $O(\log(L + 1))$ bits. This supplies a deterministic asymptotic extension; it is not implemented in the fixed palette interface. A user can also supply a sufficiently large certified finite palette for a specified input-size range.

The current code accepts distinct odd primes below 2^{31} , and defaults to (65521). It tests the explicit completion inequality after every full prime pass. If the supplied palette is exhausted before that inequality holds, the record says that threshold completion was not certified. Scanning then continues. The code neither invents missing modulus bits nor interprets an inconclusive residue as an unknot certificate.

5.3 A cost account for the implemented observer

Let N be the original number of crossings. For observed suffix s , write n_s for its crossing count, b_s for its frontier size, v_s for the number of common black-fragment vertices, ℓ_s for the number of primes processed, K_s for the number of distinct boundary matchings queried, and $D_{s,p}$ for the number of distinct black terminal partitions queried at prime p . Let S_s be the total number of represented objects traversed in the observed whole components. These are measured workload parameters, not promised small functions of N .

The main costs are as follows. Ordinary dictionary and disjoint-set operations are stated in the usual unit-cost model; field and large-integer costs are separated explicitly.

Operation	Cost and limitation
Source coloring	Once, near-linear in N , including spherical validation.
Suffix geometry	$O(N + n_s \log(n_s + 1) + v_s^2)$ elementary work is a safe bound. The current constructor revalidates the whole crossing order at each stage; it is not a suffix-only $O(n_s)$ routine.
Boundary acceptance and counts	$O(b_s \log(b_s + 1))$ per new matching is a safe bound for normalization and boundary disjoint-set summaries.
Response preparation	$O(v_s^3)$ field operations and $O(v_s^2)$ temporary field elements per prime.
New determinant query	$O(1 + b_s^3)$ field operations, using a matrix of size at most $b_s - 2$ when $b_s > 0$; immediate zero when radical dimension is too large.
Whole-component aggregation	$O(\ell_s S_s)$ four-coordinate visits, in addition to grading recovery and exact Euler accounting. No norm is applied to a proper subset of one component.
CRT and norm minimization	Four-coordinate integer arithmetic per component and prime, on integers whose bit length includes $\log M$ and the exact sums.

In particular, the determinant field-operation term is bounded by

$$O\left(\sum_s \left[\ell_s v_s^3 + \sum_{p \text{ mprocessed}} D_{s,p}(1 + b_s^3) \right]\right). \quad (20)$$

This term must be added to the scan itself, geometric validation, grading recovery, coefficient accounting, cache operations, and CRT bit costs. If prime bit lengths and the total modulus bit length are polynomial in N and $\log(S_s + 1)$, each displayed elementary arithmetic operation has polynomial bit complexity. This suffices for the conditional asymptotic conclusions below; we do not equate a Python big-integer operation with a constant-time bit operation.

Active nonzero response data use $O(\ell_s(1 + b_s)^2)$ field elements in addition to the common dense graph. The implementation also retains per-stage matching and cofactor caches, and completed value caches whose keys include earlier stages. Its *total* memory is therefore not bounded by $O(b_s^2)$. Memory accounting must include at least the recorded matching keys, four-vectors, Euler values, and the scanner's own objects and morphisms.

5.4 What a width bound would and would not buy

There are at most

$$(b - 1)!! \leq b^{b/2} = 2^{O(b \log b)}$$

perfect matchings on a labeled even frontier. Black terminal partitions are also bounded by the Bell number on $b/2$ elements, hence by $2^{O(b \log b)}$. These bounds require no single-disk assumption. If all completions are known to be noncrossing on one fixed disk boundary, a Catalan bound may be substituted; the present general suffix representation does not establish that additional hypothesis.

Consequently a polylogarithmic frontier bounds the *number of possible oracle keys* quasi-polynomially. For example, $b = O((\log N)^2 / \log \log N)$ gives at most $2^{O((\log N)^2)}$ matching keys. The stronger restriction $b = O(\log N)$ gives $2^{O(\log N \log \log N)}$ keys by this elementary estimate. These are oracle table bounds, not complete recognition bounds.

Many algebraic generators may have the same boundary matching and different differentials, gradings, or continuation behavior. Determinant caching removes repeated observation cost for that matching; it does not identify those algebraic generators. Moreover, crossing order optimization has no proved polylogarithmic-width guarantee on arbitrary inputs in the maintained program. Planarity by itself is not such a guarantee.

Proposition 5.3 (A precise conditional route to quasi-polynomial time). *Consider a family of validated diagrams with N crossings. Suppose there is an algorithm which, through a decisive marked obstruction or final rank calculation, constructs and updates the required relative complexes in $2^{(\log N)^{O(1)}}$ bit operations and space, including every represented object and morphism. Suppose also that the number of stages and oracle calls is quasi-polynomial and that exact sums and multiplicities have polynomial or quasi-polynomially bounded bit length. Then adding the deterministic modular observer with sufficient certified modulus preserves a quasi-polynomial bound.*

Proof. A suffix has $O(N)$ darts and at most $O(N)$ common graph vertices. Preparation and each determinant therefore have polynomial dimension. The coefficient bound requires $O(N + \log(S_s + 1))$ modulus bits; the hypothesis makes this polynomial in N for a quasi-polynomial number of explicit objects. The sieve construction above supplies enough primes with polynomial bit length in polynomial time in the required modulus bit length. Every observer operation has polynomial cost in its relevant bit lengths, hence at most quasi-polynomial cost in N , apart from its already counted traversal of represented objects. A quasi-polynomial number of such operations and the assumed representation work remain quasi-polynomial. $\square$

The main hypothesis is exactly the unresolved obligation. The proposition is an interface theorem explaining where a future representation result would attach, not a proof that this hypothesis holds. Even a constant-size determinant cannot pay for exponentially many distinct differential generators. Conversely,

a structural family formula may succeed despite an exponentially large explicit basis, as the three-braid literature demonstrates [6].

There is a second independent obstruction: the four-residue bound can remain one for every prefix examined. Khovanov unknot detection concerns the full homology, whereas these Euler values retain only a small part of it. A universal bound on the first decisive prefix would need a new theorem about continuation behavior. None of the modular arithmetic results supplies that theorem.

6 Implementation, replay, and correctness evidence

6.1 An optional backend with a small integration surface

The new public selector is `backend="shadow-modular"`. The existing `standard` default and existing `shadow` integer observer retain their roles. The production additions are deliberately separated by invariant:

Module under <code>fast/fastunknot/</code>	Responsibility
<code>boundary_tait.py</code>	Checked cut-face geometry, canonical black partitions, spherical acceptance, and boundary component counts.
<code>modular_response.py</code>	Prime validation, symmetric elimination with one- or two-coordinate pivots, retained radical, and quotient cofactors.
<code>modular_lattice.py</code>	CRT, balanced representatives, exact-sum minimum norm, and certified threshold completion.
<code>modular_shadow.py</code>	Whole-component observation, adaptive prime passes, resource accounting, and capped scan fallback.
<code>modular_verify.py</code>	Replay of positive raw observations using the old integer observer and an independent minimum-norm calculation.

The changes to `recognize.py` and `__main__.py` register the selector and `shadow_primes` option. The five production modules have no dependency on archived report packages or the earlier rational prototype. They use the Python standard library and the maintained package.

A normal command is:

```
python -B -m fastunknot recognize examples/conway.json \
--backend shadow-modular --shadow-primes 65521
```

The usual filters may decide this input before the selected backend runs. For a direct reproducible marked-observer example, the supplied demonstration uses the closure of $(\sigma_1\sigma_2)^5$ and the natural crossing order:

```
from fastunknot import Diagram
from fastunknot.modular_shadow import modular_shadow_khovanov_decide
from fastunknot.modular_verify import replay_modular_shadow

D = Diagram.from_braid(3, [1, 2] * 5)
claim = modular_shadow_khovanov_decide(
    D.pd, order=list(range(10)), shadow_max_work=None,
    primes=(3, 5, 7, 11, 13))
assert claim["method"] == "marked-residue-four-modular"
assert claim["stage"] == 9
replay = replay_modular_shadow(D.pd, claim)
assert replay["verified"]
```

This example illustrates a genuine proper-prefix obstruction, including a whole component with more than one object. It is not the benchmark family.

6.2 Saturation and transactional resource behavior

The mathematical L_M is the full sum in equation (13). The decision routine returns its threshold-saturated form $\min(2, L_M)$ and may stop after a subset of *whole* components already contributes at least two. It records both the observed and total component counts. A positive result certifies the Boolean comparison even if the sufficient modulus-size inequality was not needed. A negative comparison is labeled complete only after every relevant component has been processed and the coefficient bound proves completeness.

A geometry, response, cofactor, or completed vector enters its cache only after its computation finishes. Deadline or work interruption cannot publish a partial elimination as if it were a valid response. Already completed exact Euler values remain usable. Every cache hit still checks the global deadline.

The optional observer has two local allowances. Exhausting the determinant work allowance disables the modular shadow and retains the existing Euler reserve. Exhausting the shared completed-state allowance disables further observations and lets the capped complex scan continue. A geometric inherited-color decline also disables the optional representation. Global time or object exhaustion propagates through the public API as `UNKNOWN`. With no such global resource bound, the finite exact capped scan remains the fallback decision procedure. Small modular norms, zero cofactors, and exhausted prime palettes never produce `UNKNOT`.

The deadlines are cooperative: expensive Python arithmetic, allocation, and individual library calls cannot be interrupted at every machine instruction. The benchmark therefore adds a separate POSIX signal-based wall-clock cap, while the production interface preserves the established cooperative resource semantics.

6.3 What the replay verifier proves

A positive raw observation contains the source crossing order and surviving mark, the proper stage, distinct whole component identities, recovered shifts, saturated positive multiplicities, the processed primes, the exact CRT product, four residues, exact Euler sums, coefficient bounds, minimizing lifts, and claimed norms.

The verifier reconstructs the prefix using `ComponentScan(rank_cap=3)`. It validates the whole-component identities against the reconstructed differential graph. It then evaluates completions using the older `ClosureShadow` integer implementation. Thus the new boundary geometry, finite-field response, and production lattice minimizer are absent from this arithmetic verification path. It checks every aggregate residue against the independently computed integer vector and recomputes the constrained minimum by a separate marginal-cost formula.

Checking minimality is essential. A congruent vector with the right coordinate sum is a feasible lift, but its norm can be larger than the true minimum. One tampering test supplies a feasible lift of norm four where the correct modular minimum is two; replay rejects it even though four also equals the independently computed integer norm. A claimed minimum is not authenticated merely by exhibiting one feasible vector.

The verifier accepts distinct complete component records whose verified weighted sum exceeds one; other components need not be listed. It rejects partial components, duplication, false shifts, incompatible source data, composite or repeated primes, false products, incorrect residues, and nonminimal lifts. Resource exceptions propagate rather than being converted into acceptance. Runtime counters and performance metadata are outside its certificate scope.

This is *full prefix replay with an independent observer arithmetic path*. It is not a short standalone NP certificate, an independent implementation of the scanner, or a formal proof-assistant development. Replaying a large prefix may cost as much as constructing it. Those distinctions matter when using the word “certificate” in complexity claims.

6.4 Independent arithmetic and actual-diagram audits

The package supplies three new test modules, with 31 tests in total. The response tests include 1,964 randomized quotient comparisons against both dense modular determinants and integer Bareiss determinants. They force singular interiors, two-coordinate pivots, exceptional primes, loop-producing identifications, nonclassical pairing rejection, and deadline interruption. End-to-end tests check real suffix completions, mirrors, grading recovery, public dispatch, Euler fallback, closed-rank unknot decisions, and replay corruption.

A separate arithmetic script imports no production modular code. Its independent dynamic-programming enumeration checks 33,300 norm-minimization instances, and its exhaustive vector checks test 1,496,340 weighted threshold cases. These are finite audits of the implementation and statements; the general proofs remain necessary.

The larger diagram audit chooses 64 seeded random small braid inputs and five fixed inputs, then includes each mirror. Every diagram has at most nine crossings so that an independent reduced $\mathbb{F}_2$ cube is feasible. The saved input list and all crossing orders make the result reproducible without repeating the random selection.

Table 1: Reproducible audit against independent reduced cubes and integer completions.

Check	Count
Diagrams, including mirrors	138
Independent reduced $\mathbb{F}_2$ cubes	138
Cube generators and checked d^2 columns	209,658
Proper prefixes	868
Actual matching completions	2,516
Modular vector comparisons over six primes	15,096
Bounds compared with full reduced homology	868
Certified adaptive threshold agreements	868
Final capped ranks and recognition statuses	138
Positive modular claims replayed	50

All these comparisons pass. The cube code comes from report 26 and constructs its own differential and homology; it is independent of the maintained scanner’s cancellation machinery. The audit explicitly checks $d^2 = 0$. It compares both proper-prefix lower bounds and final capped ranks with the independent reduced homology.

Prime three alone misses 70 real obstructions in this audit. It produces no false positive. The adaptive supplied palette resolves every one of the 868 threshold comparisons in the audit. This is direct evidence for the one-sided meaning of modular observations and against treating an inconclusive small prime as an unknot test.

The geometric audit observes a maximum frontier of 14 darts, seven terminals, residual nullity two, and actual query dimension five. It checks $t = b/2$ on 730 nonempty-frontier preparations, the $b - 2$ bound on 14,898 potential queries, and 198 radical-dimension zero cases. These include singular cases absent from the timing family.

6.5 Full-suite status and an inherited limitation

A broader run of the maintained test suite discovered 708 tests, including the first 21 new modular tests. It recorded 704 passes, three optional-Regina skips, and one error in an existing normal-surface subprocess test. The ten replay tests were added afterward and are included in the separate complete 31-test modular run.

The error reproduces both in isolation and against the untouched baseline `normal_surface.py`. On this CPython 3.12.14 POSIX runtime, a timed `communicate(payload)` call can leave input pending; retrying with `communicate(None)` retains the internal buffer but no longer schedules stdin writes. In the test, 65,536 of 500,000 bytes are sent and the child waits for the remainder until the local deadline. A single uninterrupted communication call completes successfully. The package records this as a real inherited optional-worker I/O defect, not as benchmark contention or a passing gate. No source change in that unrelated worker is included here.

The failure limits an optional native-worker path and does not invalidate the modular arithmetic comparisons. It must nevertheless remain visible to anyone reproducing the full suite. A separate repair needs to preserve pending-stdin progress, global cancellation, and child cleanup together.

7 Experiments: arithmetic gains and recognition regressions

7.1 Protocol and comparison targets

The experiment distinguishes three scopes: repeated observations of one actual suffix, complete raw shadow scans, and the public recognition pipeline. Mixing these scopes would obscure both the improvement and its limitations.

For repeated observations, the competitors are the maintained direct integer `ClosureShadow`, the earlier rational `determinant_research/boundary_tait.py` prototype, and the new modular observer with prime 65521. An identical second integer arm controls for ordering and timing variation. Every arithmetic call constructs a fresh engine and includes source geometry, response preparation, query processing, and result collection. No response is warmed outside the timed region.

Each input has one excluded warm-up followed by five rounds with shuffled arm order. The data retain every elapsed time, arm order, result, input diagram, crossing order, actual matching stream, source hash, and censoring flag. Very short pipeline calls use repeated fresh instances and record the individual call times before forming the round average. Paired speed ratios are formed *within each round*, then their median is reported. They are not the ratio of two separately rounded medians.

Each call has a three-second cooperative allowance and a separate 3.15-second POSIX signal-based wall-clock cap, implemented with `SIGALRM` and `setitimer`. The environment is CPython 3.12.14 on Linux x86_64; the raw file records the full platform and interpreter strings. Five repetitions support a local performance comparison, not a distributional or asymptotic claim. Intervals in the figure are the observed minima and maxima of the five paired ratios, not confidence intervals.

7.2 A controlled family of actual query streams

The base input is a five-strand braid with signed generator indices

$$(-2, 3, 2, 2, 4, 2, -3, -2, -1, 1, -2, -1, -2, -2).$$

Its base crossing order is

$$(5, 1, 10, 2, 9, 11, 13, 6, 7, 0, 12, 4, 8, 3).$$

Append k copies of

$$(1, 1, -2, -2, 3, 3, -4, -4)$$

and place their crossings after the base order. This appended braid is pure: each generator occurs in an adjacent square, so its permutation is the identity. The closure therefore keeps the base knot's one-component permutation. The actual planar diagram is validated as usual.

After the first nine crossings, collect the 200 distinct matchings present in the real prefix scan. No arbitrary set partitions or invented morphism graph is substituted for them. This keeps the observation stream tied to the actual algorithm while increasing the size of the unprocessed interior. The sequence has $n = 5 + 8k$ suffix crossings and the dimensions in table 2.

This family is a controlled arithmetic workload. It is not claimed to be a hard family for complete unknot recognition, and the fact that its appended braid is pure does not say that its knot type is unchanged. The fixed source, not a knot-equivalence claim between rows, defines the experiment.

7.3 The repeated-query improvement

The all-terminal row has no interior work to amortize, and the modular observer is slower than both predecessors. At three interior vertices it beats the maintained direct observer but is still slower than the rational prototype. At 15 interior vertices the rational and modular medians are close. At 47 interior vertices, the modular path gains a median factor of 26.666 over the direct integer path and 3.968 over the rational prototype. At 127 interior vertices, the 50-query workload gives factors 10.743 and 10.218 respectively.

Table 2: Actual repeated-query suffix dimensions.

k	n	Queries	b	v	t	$v - t$	Cofactors
0	5	200	18	9	9	0	61
1	13	200	20	13	10	3	50
4	37	200	20	25	10	15	50
12	101	200	20	57	10	47	50
32	261	50	20	137	10	127	12

Here k is the pure-tail repetition count, n the suffix crossings, b the frontier darts, v the common black-fragment vertices, and $t = b/2$ the terminals. Cofactors counts distinct terminal partitions actually evaluated. The last row intentionally uses only the first 50 of 200 available matchings.

Table 3: Fresh-setup repeated-query times and paired speed ratios.

k	Integer (s)	Rational (s)	Modular (s)	Int./mod.	Rat./mod.
0	0.008971	0.007237	0.011583	0.670	0.608
1	0.022148	0.006680	0.011787	1.878	0.573
4	0.088640	0.014030	0.013336	6.499	1.016
12	0.734303	0.100374	0.027537	26.666	3.968
32	1.488842	1.428603	0.138268	10.743	10.218

Times are medians of five complete rounds. Ratios are medians of within-round ratios, so they need not equal ratios of the displayed medians. A ratio above one favors the modular evaluator. Prime 65521 is used throughout.

The identical-control median ratios in this arithmetic experiment range from 0.953 to 1.036. This supports attributing the larger favorable ratios to a real change in arithmetic work rather than simply to the order in which the arms were timed. It does not turn the small-instance ratios into precise universal constants.

The final row requires a censoring explanation. An initial 200-query pilot at $k = 32$ exhausted the three-second allowance for both direct-integer arms, while the rational and modular arms completed. That pilot is retained in the raw record, but no ratio is inferred from its censored times. A separate explicitly shortened first-50-query workload then completed in all arms and supplies the reported five-round row. Comparisons within that row are valid; its setup is amortized over fewer queries than in the preceding rows.

All timed modular interiors have nullity zero. These measurements therefore support amortization and modular-arithmetic performance, not singular-case performance. Singular correctness is established by the algebraic and actual- diagram audits in [section 6](#).

7.4 The full scans are slower on this corpus

The seven examples cover the Conway knot, the Kinoshita–Terasaka knot, visible connected sums of the Conway knot, two unknot examples, and a five-strand stress braid. The example names identify the saved input files or generators in the package; they are not a random sample of knots at those crossing numbers.

Every raw-scan median ratio is below one. The observed range is 0.536–0.987, with the largest relative regression on the short stress-braid scan. Calling through the public recognition pipeline with its preceding filters disabled gives the same qualitative result, with median ratios 0.515–0.969. These absolute runtimes are short, but the experiment provides no evidence that the modular backend improves complete scans on this corpus.

The normal pipeline is a separate and especially important control. All seven examples are decided before either shadow observer is invoked: Conway and Kinoshita–Terasaka by the modular Jones filter; their visible sums by factorization followed by that filter; the hard-unknot example by the three-braid free-product route; the scrambled grid unknot by Reidemeister reduction; and the stress braid by the modular Alexander filter. Near-one ratios in those runs measure front-end timing variation. They must not be reported as evidence for the new continuation.

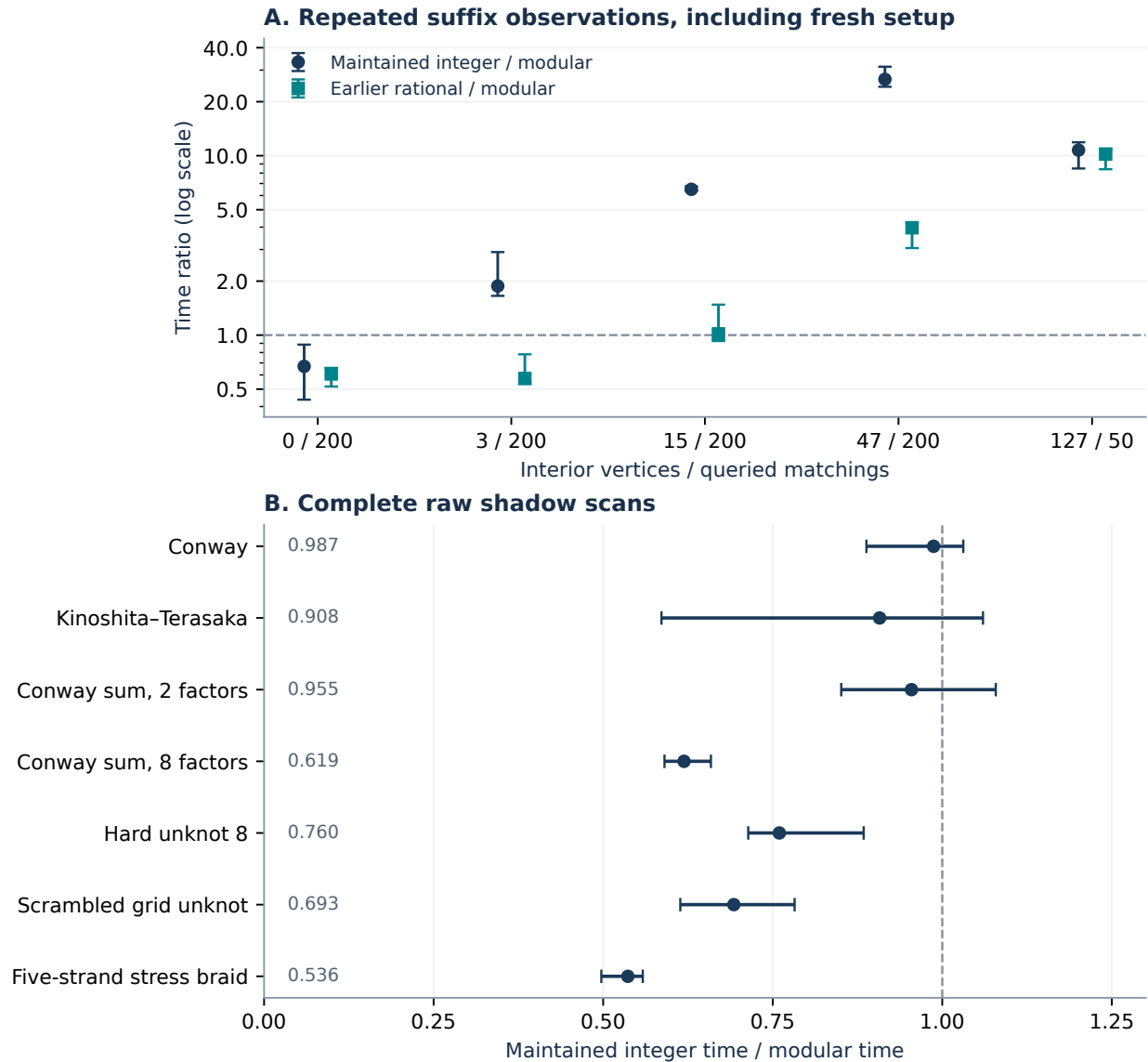


Figure 1: Measured gains depend strongly on scope. A ratio above one favors the modular implementation. Points show medians of five paired ratios; whiskers show the observed range. Panel A includes fresh geometry and response setup. Panel B measures complete raw scans, where all seven median ratios are below one. The two panels use different axis scales.

Table 4: The raw scans do not show an overall recognition gain.

Input	Integer (ms)	Modular (ms)	Raw ratio	Filters off
Conway	4.774	4.895	0.987	0.969
Kinoshita–Terasaka	5.396	5.737	0.908	0.873
Conway sum, 2 factors	8.807	9.494	0.955	0.961
Conway sum, 8 factors	10.113	16.186	0.619	0.635
Hard unknot 8	0.972	1.264	0.760	0.803
Scrambled grid unknot	0.513	0.741	0.693	0.677
Five-strand stress braid	1.186	2.158	0.536	0.515

The first two columns time raw shadow scans. Raw ratio is the paired integer/modular ratio in that scope. Filters off is the corresponding ratio through the public recognition pipeline with its earlier filters disabled. All seven ordinary default-filter runs stop before either shadow backend.

Across all warm-ups and samples, 24,200 repeated vector comparisons and 2,013 repeated status comparisons agree. These are validation-event counts, *not* counts of distinct diagrams or completions. Completed integer and rational vectors agree with the new vectors modulo 65521. Status equality is required; equality of every stopping stage or intermediate method is not. The hashed observer sources did not change during the benchmark.

7.5 The supported performance conclusion

Boundary reuse removes repeated dependence on a large suffix interior, and finite-field arithmetic can improve substantially on the rational response when that interior is large. The production measurements justify keeping this capability available. The complete-scan results simultaneously show that eager preparation is not a successful universal policy. A scheduler must estimate whether a suffix will receive enough distinct useful queries before the scan moves on or another obstruction succeeds.

The next benchmark should hold out a corpus where the earlier structural, Alexander, and Jones filters genuinely remain inconclusive, include large interior graphs with few queries, and measure successful early stops and total scan work. A claimed default-policy improvement should survive that test with setup, failed probes, prime refinement, cache memory, and timeouts all charged.

8 Further research questions and proposed priorities

The arithmetic improvement is useful enough to integrate as an optional capability. The evidence suggests a different order of priorities for obtaining practical speed-ups and for proving a quasi-polynomial theorem. The former needs better scheduling and reuse; the latter needs a representation theorem and a bound on decisive work. The following questions state both the proposed work and the evidence that would count as progress.

8.1 Near-term implementation and experimental questions

Q1. When should a suffix pay for a response? Can a scheduler predict the number of distinct useful matching and partition queries before the suffix changes? A candidate decision rule should compare measured direct-query cost with preparation cost plus expected boundary-query cost, using $v - t$, t , projected query count, and the number of likely prime passes. It should also account for the chance that the exact Euler prefilter or first few direct queries stop the scan. A concrete milestone is an opt-in hybrid that selects direct integer, rational response, or modular response on a held-out corpus and improves total recognition time with every failed probe charged. The current seven raw regressions make this the first performance priority. A model fitted only to the favorable pure-tail stream would not be adequate evidence.

Q2. Can a response survive a change of suffix? Removing one crossing changes both edges and the cut-face fragmentation. It can split a region, create new terminals, and change the radical dimension. A correct incremental update therefore needs a precise map between successive vertex spaces, not just a rank-one edge update to a fixed Laplacian. Develop an update calculus for fragment splits, terminal promotions, and edge deletion, with a rebuild path when the pivot profile changes. The useful theorem would bound amortized update work in terms of the changed frontier neighborhood. The useful experiment would include many consecutive observed stages with few queries per stage, where full rebuilding currently loses.

Q3. How should articulation blocks and sparse elimination combine? The maintained direct observer already factors at articulation points. A boundary response could eliminate terminal-free blocks separately and retain small interfaces at blocks that contain frontier terminals. The challenge is to prove the gluing formula with signed weights, zero tree factors, and singular interiors; a division by an apparently harmless block determinant can fail modulo a prime. Establish a singular-safe block response and compare it with both existing articulation factoring and dense modular preparation. Measure fill-in, temporary memory, and cost at exceptional primes, as well as elapsed time on nonsingular examples.

Q4. Which certified coefficient bounds repay their cost? The bound $C_a \leq s_a 2^{n-1}$ is universal but can be loose. Candidate refinements include the unsigned spanning-tree count of the actual graph, products of blockwise bounds, bridge reductions, and degree-based determinant bounds. A bound should be computed once and reused across completions when possible; otherwise its cost can exceed the modular observation it was meant to save. The target is a proved C'_a together with a measured reduction in prime passes or exact-threshold completion time. Signed cancellation may improve the actual coefficient, but cannot justify shrinking a certified bound without evidence.

Q5. Can prime scheduling exploit the exposure theorem? Proposition 4.8 bounds how many primes can hide an existing obstruction. Use it to design a policy choosing between one more independent prime, another CRT pass, and continuing the scan. An asymptotic version should supply enough primes automatically and charge their generation and validation. A practical version should distinguish a large default prime from a random prime-pool experiment, retain monotone CRT information, and report the cost of inconclusive passes. Prime choice must never change the meaning of a positive result. The 70 misses at prime three are a useful adversarial check, not a reason to label the method probabilistic or unsound.

Q6. What is the true memory cost of continuation reuse? Only the active residual matrices are boundary-sized. Completed-value keys and records can accumulate across stages, and algebraic generator storage may dominate everything. Introduce a memory ledger for geometry, prime responses, partition caches, matching keys, exact Euler integers, and scanner objects. Study bounded caches or stage-local eviction with reconstruction charged. A credible performance report should include peak memory and the number of retained objects, not only the largest query matrix. This may reveal a regime where recomputation is preferable to retaining all exact completion values.

8.2 Questions that address the quasi-polynomial barrier

Q7. Which continuation behaviors admit a succinct quotient? Equality of boundary matchings is too coarse to identify chain complexes with different differentials. Seek an equivalence relation on relative complexes that preserves the needed threshold decision under all admissible future crossings or actual suffixes, and that is stable under tensoring, delooping, and cancellation. The essential result would bound the number and bit size of its states by $2^{(\log N)^{O(1)}}$ and give constructive transition rules. A finite table of scalar Euler observations alone is unlikely to determine all future homological behavior; this should be tested with counterexamples before adopting it as a state representation. Any successful quotient must retain enough information to justify the marked homotopies, not merely $d^2 = 0$.

Q8. Is there a family theorem for the first decisive observation? For a specified diagram family and order policy, can one bound the number of represented objects and stages before $B_4 > 1$, or before a rank-two contraction is exhibited? The arithmetic completion theorem starts only after an exact obstruction is present. It says nothing about when that happens. A useful intermediate target is an explicit family beyond the already solved structural cases, with a proved stopping-stage bound and a certified complexity expression in its parameters. Failed examples should retain the exact sequence of prefix bounds, so a proposed detection lemma can be falsified by genuine complexes.

Q9. Can larger residue systems remain efficiently observable? Residues modulo eight or higher powers of two refine quantum grading and can retain cancellations lost modulo four. Investigate whether their evaluations admit a manageable transfer representation on bounded interfaces. The special value at i has a spanning-tree determinant formula; higher roots do not inherit that formula automatically. A successful extension needs three proofs: that the residue norm still bounds completed homology, that the relevant integer or cyclotomic data are computed exactly, and that their observation cost has the claimed dependence on boundary width. A larger invariant that requires exponential work in the suffix would not resolve the present barrier.

Q10. Can homological information be compressed without changing fields? The auxiliary primes in this paper reduce integer Euler data. They cannot be used to compute a generic odd-characteristic Khovanov rank and silently treat it as the rank over $\mathbb{F}_2$. Investigate compact descriptions of persistent cancellations, module decompositions, or continuation-compatible rank defects *over the actual coefficient field*. The maintained interval and scalar- splitting backends provide concrete comparison points. A new theorem should state exactly which operations preserve the compressed description and what happens when that description must expand. The three-braid structural formulas suggest that family-specific symbolic answers can be much smaller than explicit bases [6].

Q11. What geometric decomposition gives the right width measure? A polylogarithmic number of frontier darts would bound the oracle table, but it is not guaranteed by the current crossing-order search. Investigate hierarchical decompositions whose interfaces carry both the tangle data and sufficient algebraic summaries. A useful theorem must control interface size, number of pieces, transition cost, and the growth of multiplicities and coefficient descriptions simultaneously. If only a planar separator estimate with a much larger boundary is available, state the resulting subexponential bound honestly; it does not become quasi-polynomial through terminology. Connections to hierarchy-based topological methods are promising only after the translation cost is made explicit.

Q12. Can independently verified compressed contractions replace replay? The current verifier reconstructs the entire prefix. A more useful certificate would encode contractions and direct-sum decompositions in a form whose local identities can be checked without expanding every intermediate complex. Study straight-line descriptions of maps and homotopies, canonical sharing, and certified transfer across a decomposition tree. The certificate must include its own size bound and account for zero tests and equality of compressed maps. A short list of moves is insufficient if one move refers to an exponentially large object. A constructive bound here would address both practical replay cost and a major part of proposition 5.3.

8.3 External algorithm audits and integration milestones

Two literature tracks deserve separate, explicit audits. Lackenby’s hierarchy program supplies a geometric framework and efficiently checkable objects; an implementation study should identify which construction and encoding bounds are already proved and which speed-up lemmas remain proposed [7, 8]. Musick’s September 2026 polynomial-time claim should be assessed independently: formalize its state representation, verify that every claimed local move is discoverable within the compressed encoding, and prove a bound on the total descriptions created, including overwritten or retained grammar versions [10]. This article neither adopts

that claim nor supplies a counterexample. Its relevance is that a serious current research program should investigate a directly competing claimed algorithm instead of declaring it absent.

For ProveIt, the immediate integration sequence is concrete. Add the optional backend and its replay tests against the pinned source. Preserve the recorded arithmetic and actual-diagram audits. Keep the raw timing samples and the negative complete-scan results next to the favorable response experiment. Then evaluate a scheduling change as a separate patch with a new held-out corpus. The inherited normal-surface I/O defect should also receive a separate focused repair rather than an unexplained test exclusion.

The next mathematical milestone should be a family theorem controlling the size of a continuation-compatible representation through a decisive event. The modular response provides a proved inexpensive observation interface to attach to that theorem. Establishing the representation and stopping bounds is the work that would turn the present performance improvement into progress on a general quasi-polynomial recognition guarantee.

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